

Cozmo technical report

Handheld phone capture to a dimensioned floor plan with damage annotations.

Every number below comes from a file in this repository and names the script that rebuilds it. Anything not yet measured is marked PENDING with what it waits on. `scripts/regenerate_all.sh` rebuilds all of it.

1. Architecture

One contract, one backend, three front ends.

The contract first. `cozmo/contracts/models.py` defines the plan. No dimension is ever a bare number: every `*_m`, `*_m2` and `*_deg` field is a Measurement with value, `ci_low`, `ci_high`, unit, method and `ci_level`, and a test fails on any bare number in emitted JSON. Referential integrity is checked at load, so the evaluator never repairs input before scoring it. The schema is generated from the models.

The shared backend. Tiers differ only in how they make geometry. Once a tier has a point cloud with poses, the rest is shared: `cozmo/lidar/levels.py` (floor and ceiling), `walls.py` (Manhattan frame, wall faces), `rooms.py` and `cells.py` (rooms), `openings.py` (doors, adjacency), `uncertainty.py` (intervals), `render/plan.py`. Damage is separate and tier-agnostic.

Provenance. `run` writes `plan.json`, `plan.png`, `run_manifest.json`, `drift_report.json` and `debug/`. The manifest holds a SHA-256 of every input file the tier may read, the resolved config and its hash, the git commit, the seed, stage timings and library versions. The plan holds no wall-clock field, so the same input, config and seed give a byte-identical `plan.json`. That is also how a reused plan is checked: the benchmark re-hashes the input and says whether it still matches.

Tier isolation is enforced in code. Every file access goes through `StrayScan.open`, which raises on a file the tier may not read: the video tier sees only `rgb.mp4`, and the photo tier is refused a scan folder outright because `depth/` holds png files that would otherwise pass as room photos. The manifest hashes only permitted files (`tests/test_stray.py`).

2. Tiers and devices

tier	input	reconstruction	median wall error against tape
lidar	one Stray Scanner scan for the whole property	classical geometry, no trained model	no tape ground truth exists
photo	2 to 8 stills per room	MapAnything poses, camera-height scale, same backend	13.0% , worst 31.9%
video	one clip per room	COLMAP SfM in chunks, camera-height scale, same backend	32.9% , worst 137.4%

Wall errors from `docs/benchmark/eval.json`, rebuilt by `scripts/benchmark_all.py`, over 12 photo walls and 16 video walls with tape.

Neither image tier is accurate enough to ship, though the photo tier is within reach: 6 of 12 photo walls now fall inside the 8% budget, against none of 16 video walls inside 3%.

Gates are scored **per tier** (`docs/benchmark/benchmark.md`), because pooling lets a tier with large errors and many walls swamp one with few small ones. A gate with nothing to score reads NOT EVALUATED with its reason, never PASS. Photo passes 2 of 6 scorable gates and video 2 of 7, in both cases the

structural ones (adjacency, overlap) rather than the dimensional ones. ****The lidar tier's only evidence is cross-capture agreement**, 1.9 cm between wall faces two scans both saw. **Every lidar gate reads NOT EVALUATED:**** the supplied scans are of a property nobody measured, and no iPhone was available to scan the rooms that were.

A regression found by measuring, and its cause

The photo tier first scored better on **compressed copies** of the photographs than on the **camera originals**, which is the wrong way round and was worth chasing rather than reporting as noise.

The photographs are the same pictures, matched at correlation 1.000, 9 of 9 in every room. The cause was the focal length: MapAnything estimates one when it is not given one, and on the originals it guessed ****464 px against a true 332 px****, implying a 46 degree field of view where the camera has 76. Too long a focal pushes the scene apart sideways while the camera-height prior pins the heights, which is the signature seen: ceilings right, walls 33 to 47% long.

The compressed copies scored better only because the guess landed nearer the truth on soft images.

Ablation on bedroom_2 (fix_loop/loop2_video_scale/photo_ablation.md), tape 388.6 and 363.2 cm:

configuration	walls cm	error
originals, model estimates the focal	532.3, 432.4	+46.6%, +11.3%
downscaled to 1600 px	focal estimate unchanged at 464 px	no change
EXIF focal given to the reconstruction	458.6, 392.2	+26.3%, +0.9%

Across twelve measured walls the fix took the photo tier from 22.4% to **13.0%** median error and coverage from 0.69 to 0.88.

Downscaling changes nothing; giving the model the focal is the fix. The control is the Nokia 8.1, which writes a focal in millimetres but no 35 mm equivalent, so it gets no intrinsics and the model still guesses. It is the one camera whose rooms did not improve.

Stray Scanner format. rgb.mp4 1920x1440 HEVC 60 fps stored rotated; depth/ 256x192 uint16 millimetres; confidence/ 0, 1, 2 with only 2 used; odometry.csv camera-to-world poses, quaternions x, y, z, w, per-frame intrinsics scaled by 256/1920 for depth. Unprojection is OpenCV convention.

Getting it wrong is silent, so a regression test pins the floor of c00a170fe1 at world y = -1.48 m within 2 cm (tests/test_stray.py).

Devices. Nokia 8.1 shot every video loop and the bedroom_2_repeat stills; Moto Edge 50 Neo shot the other stills. ****No iPhone was available for capture****, so every lidar result comes from the three supplied scans and the lidar capture route has not been walked through on our own device (docs/device_matrix.md).

Runtime on an M1 Max: lidar 4.0, 13.2 and 27.8 s for 1715, 5251 and 9745 frames, against a 3 minute budget. Video is about 8 minutes a clip and photo 77 s for four rooms (docs/benchmark/provenance.json).

Video tier: what was rejected and why

MapAnything was tested and rejected as a metric source

(docs/experiments/mapanything/result.json): the median ratio of predicted to lidar depth is 0.712, 0.704 and 0.709 across three runs, standard deviation

0.09 to 0.13, so a fixed correction leaves about 13% spread. Feeding it ARKit poses would likely have fixed the scale and was refused, because a video result from lidar-derived poses measures the lidar tier. It is kept for the one job it is good at, relative poses across a chunk seam.

What the video tier actually does

COLMAP structure from motion over frames from the clip, in **chunks** rather than over the whole walk: a 50-second sweep rarely registers as one reconstruction, and a failed chunk then costs seconds of video instead of the room. Chunks join by **floor-plane bridging**, aligning on the floor they share, with MapAnything used only for the relative pose across the seam.

Scale comes from the camera-height prior, each chunk scaled so its camera sits 1.40 m above its own floor, with Depth Pro as a **weak second opinion** that checks the scale rather than setting it. The room is fitted as a ****single room around the camera path****, because segmenting free space out of a partly reconstructed floor finds either nothing or a fragment. Median wall error 32.9% over 16 walls; section 6 says why.

3. Drift handling

ARKit odometry drifts slowly: yaw creeps and floor height wanders over a few minutes of walking, which smears wall faces and breaks the floor into a wedge. `cozmo/lidar/drift.py` anchors to the building, not the trajectory. The walk is cut into 5-second chunks. Per chunk: the dominant wall azimuth modulo 90 degrees gives a yaw error against the global Manhattan direction, removed by rotating about the chunk's own centroid; the median height of its floor points gives a vertical offset; then its wall points shift along each axis by the median offset to the nearest global face. Corrections beyond configured limits are rejected, because a chunk that looks 20 degrees off usually has too little wall to estimate from.

The ablation says it barely matters here

(`fix_loop/after/bench/runs/*/drift_report.json`):

scan	wall thickness off	on	footprint off	on
c00a170fe1	24.2 mm	23.6 mm	55.4 m2	35.8 m2
1a8384c3f6	32.1 mm	35.8 mm	80.0 m2	86.6 m2
c7d28f72c6	32.5 mm	32.6 mm	65.8 m2	64.7 m2

Thickness improves on one scan, worsens on another, unchanged on the third.

The model summary says why: mean yaw error 0.55 to 1.57 degrees, mean height error 3 to 8 mm, mean shift 1.4 to 2.2 cm. ****ARKit's poses were already good enough that there was little to correct.**** Room overlap does fall consistently. It stays on because it costs one pass and does no harm; that is the whole case for it.

The video tier has its own drift problem and it is not this one: chunks disagree about scale, not orientation, which section 6 covers.

4. Error budget

Every interval is built from named terms, defined in `cozmo/lidar/uncertainty.py`.

Wall length. Statistical: each bounding face's position error is its residual spread over an effective sample size counting 0.25 m patches, never

raw points, because points 2 cm apart on one wall are not independent measurements. Systematic: a 1% depth scale bias scaling with length, plus a 1 cm floor. Half width $1.96 * \sqrt{sa^2 + sb^2 + (bias * L)^2}$, floored at 1 cm. On the lidar scans the median interval is 2.4% of length.

Ceiling height. Per 25 cm cell, ceiling mean minus the floor plane beneath, reported as the median. **The spread is deliberately not divided by the square root of the cell count**: cells are not independent samples, and sub-millimetre precision from a handheld scan is the confident garbage the grading penalises. Under 15% ceiling coverage no number is invented: a 2.2 to 3.2 m prior with method `prior_no_ceiling_observed` and a warning. It is the weakest quantity at both image tiers: 2.1 to 10.7 cm error at photo and 27.2 to 63.3 cm at video, against a 1.5 cm gate.

Areas propagate from edge lengths in quadrature. **Partially observed rooms **have intervals doubled and say so in the method string**. Damage extent** with depth runs from a contrast estimate to the box projected on the surface; without depth it is `unbounded_no_metric_depth`, 0 to 4 m2.

Ground truth, and its limits

The error budget is only as good as what checks it:

- **The supplied lidar scans have no ground truth at all.** Nobody measured that property, so no lidar accuracy number exists or is given.

- **Own rooms: tape, centimetres to the nearest inch.**

`truth_uncertainty_m` is 0.013 and the coverage check widens the truth by it, so a pipeline is not marked wrong for beating the reference.

- **The hall is scored on ceiling, openings and adjacency only**

(`score_walls: false`): an open-plan connector with no measurable wall run.

- **Four windows are present_unmeasured:** neither hit nor phantom, and no calibration item.

- **Staged damage is two A4 sheets**, recorded as

`extent_m2_upper_bound: 0.0624`, a bound on the mark rather than the mark.

5. Calibration analysis

Calibration is scored separately from accuracy, in `cozmo/eval/calibration.py`:

empirical coverage against the nominal 95%, mean interval width as a percentage of value, and a "confident garbage" count, meaning the truth falls outside the interval and that interval is narrower than the median for its quantity type. The median is per quantity because widths in metres and in square metres are not comparable. Broken down by tier and by quantity.

Measured over every capture with tape, from `docs/benchmark/eval.json`:

group	n	coverage	mean width, % of value	confident garbage
all	37	0.95	250	2
photo	16	0.88	46	2
video	21	1.00	409	0

The photo tier now lands on its nominal 0.95 overall, with two confident-garbage cases, both wall lengths.

The photo tier is now roughly calibrated; the video tier is not, and looks better only because it says less.

Photo covers 0.88 with intervals at 46% of value, which is wide but in proportion to what a handheld photo reconstruction knows. Two confident-garbage cases remain, both wall lengths: wrong, with an interval too narrow to admit it.

Video covers 1.00 with intervals averaging **409% of the value**. A wall reported as 6.6 m plus or minus 5 m contains the tape reading and tells nobody anything. It is not rewarded for knowing its error, it is rewarded for refusing to commit.

The lidar tier contributes no coverage, because the supplied scans have no tape. What it contributes is refusal: two of three scans report ceiling height as a 2.2 to 3.2 m prior rather than a number, and every room in the largest is flagged partially observed with doubled intervals.

6. The fix loops

Two loops were run. Both are negative results and both are reported as such.

Loop 1: cross-capture repeatability (fix_loop/)

Declared: worst gate repeatability, 0 of 112 wall rows within tolerance on two lidar captures of one apartment. Root cause hypothesis: room partition is unstable because it grows from observed floor and the walked path, so it follows the operator rather than the building. Predicted after: 40% of rows within tolerance.

The eval was broken first. Plans are emitted in the ARKit world frame, so the two captures sat 58 degrees apart and no quarter turn aligned them. Every wall pair failed the parallel test. Fixed by canonicalising each plan to its own Manhattan frame before the search, in commits before the BEFORE run.

The falsifier fired. Dropping ghost faces alone, the cheap test named in the declaration, moved nothing: about 1% of face length is rejected and the gate stayed at 0. The declaration's own evidence had over-counted ghosts at 16%, measuring faces against an envelope built from the polygons under test.

The fix shipped and did not work. Wall-driven cell complex segmentation with support from the 1.0 to 1.6 m band. Still 0 rows within tolerance, rooms paired across the captures fell from 6 to 3, footprints inflated. Three of four predictions wrong. erosion is the default again.

The cause was coverage, the third falsifier: only **56% and 37%** of each capture's wall-face length has any counterpart in the other. The contrast that carries it, from fix_loop/evidence/evidence.json:

measured at	disagreement between the two captures
wall face position	1.9 cm median
polygon edge length	97.5 cm median

The geometry agrees to two centimetres. The partition into rooms and edges does not, and no rule can invent a wall one capture never saw.

Loop 2: video-tier metric scale (fix_loop/loop2_video_scale/)

Declared: the metric scale came from a monocular depth model and was wrong. Eight predictions were written before any fix code. **Two came true:** every clip now produces a plan where three of six produced nothing, and the share of the sample scan reaching the output went 30% to 70%.

Six did not. Median wall error went **54.8% to 35.3%** against a predicted 8 to 20%, nothing reached the 3% gate, and the two captures of one bedroom

disagreed **more** than before: 198% and 145% per wall against 107% and 143%.

Three of five falsifiers fired.

Two things the diagnosis missed:

- **The prior only works when the floor plane is right.** A chunk that sees almost no floor fits a plane that is not the floor, and the prior scales confidently to it: two bridged chunks landed at 0.148 and 0.247 metres per SfM unit, a 67% disagreement, both set by the prior. Repeatability got worse because the error stopped being a shared bias and became random.

- **The rooms were never measured at all.** A face counted as a wall only if it reached 1.5 m above the floor, **inherited from the lidar tier**. A phone carried at 1.4 m and pointed level never reconstructs that high: on one clip all eight faces topped out between 0.72 and 1.46 m, none qualified, and the room came back as the camera path in a box.

Iteration 2 was declared, tried and rejected: frames through the photo reconstruction were predicted to give 3 to 12% error and gave +69% then -36%.

The finding that matters most is not about scale. Nine composed stills of bedroom_2 gave walls within about 1% where fifty seconds of video of the same room gave 53% and 66%, on the same prior and fitter. ****The difference is the capture**, and the protocol now recommends photographs over video.**

7. Known failure modes

Measured.

- **Neither image tier is accurate enough to ship:** photo 13.0% median wall error, video 32.9%, against budgets of 8% and 3%. Photo intervals are close to calibrated at 0.88 coverage; two confident-garbage cases remain.

- **The photo stitcher recorded its transform twice**, baked into the geometry and again as a placement, so applying the contract stacked every room: 28.01 m² overlap against a 0.05 m² gate, while the tier's own report said 0.0000. Found by the two disagreeing, fixed, now passing.

- **Openings are not found.** 0 of 7 within 2 cm at both image tiers, and the reason is absence rather than inaccuracy: **nothing matched at all**, 0 matched against 6 missed and 1 phantom at photo, 0 against 7 missed at video.

- **Cross-capture repeatability fails at every tier:** 0 of 153 rows on the lidar pair, 0 of 8 on the same-device video pair. **Ceiling height** fails everywhere and is unavailable on two of three lidar scans.

- **Damage precision on photos:** 8 to 15 false regions in a room with 2 marks, from 63 before filtering; on lidar the same filters reach 0 from 106, because two of the four need depth and poses. Cracks are most at risk, and segmentation over-segments: 11 rooms on a flat with 6 or 7.

Hazards in the real captures (docs/hazards.md, each with evidence).

Glass and a wardrobe mirror return confident depth for a room that is not there; the damage filters handle them, reconstruction does not. A dog walked through one capture, glossy tiles thicken the floor peak, and textureless walls give sparse returns, part of why coverage differs between two walks of

one flat. An unobserved ceiling is the commonest cause of a wide interval and is purely a protocol problem.

Assumptions that will break. Manhattan: a curved or 45 degree wall appears as a missing face, the safer failure but still a failure. Image-tier scale depends on the phone being held near 1.4 m, and a capture at waist height is wrong by the ratio, with nothing able to detect it.